NEWS-DRIVEN FINANCIAL PREDICTION ENGINE

Master Architecture Blueprint

Nifty 50 | Multilingual NLP | T-LSTM Hybrid | 3-Tier Adaptive Learning

Version 1.0 — March 2026 — Architecture: COMPLETE

Component	Status	Machine	Priority
T-LSTM Dual-Stream	DESIGNED	Machine 1	P0
FinBERT English NLP	DESIGNED	Machine 2	P0
MuRIL / Hindi NLP	EVALUATE FIRST	Machine 1	P0
CSV Handshake	DESIGNED	All Machines	P0
3-Tier Feedback Loop	DESIGNED	Machine 1	P1
Daily Scheduler	DESIGNED	Machine 3	P1
Prediction Dashboard	TO BUILD	Team	P2
Year 20 Holdout	SEALED	Machine 1	P0

1. Project Overview & Final Decisions

Objective

Predict 10:00 AM next-day price move for 10-15 Nifty 50 stocks by fusing multilingual news sentiment (Hindi + English) with 20 years of historical OHLCV data. Output is a regression float with confidence interval.

Output Formats

- Predicted % move with 95% confidence interval (e.g. +1.8% +/- 0.4%)
- Direction badge (UP / DOWN / NEUTRAL) with magnitude label
- Team dashboard: daily predictions + gap trend tracking
- predictions.csv exported each trading day, Git committed

Locked Design Decisions

- Target price: 10:00 AM (post-opening noise settlement)
- Label type: Regression float — NOT binary classification
- Label correction: Subtract Nifty 50 overnight return
- Price data source: Kaggle 20-year Nifty 50 OHLCV dataset
- Validation: Walk-forward CV + Year 20 sealed holdout
- Hindi sparse stocks: Weight English 100% (< 2 articles/3 days)
- Hindi NLP: Evaluate MuRIL vs IndicBERT in Stage 0
- Loss function: Huber loss (delta = 0.01)
- Primary KPI: MAE on predicted % move

2. System Architecture — T-LSTM Dual Stream

2.1 Stream A — Multilingual NLP Pipeline (Daily, Machines 1+2+3)

Runs each evening. Fully independent of LSTM training. Converts raw news into sentiment scores saved to the CSV handshake file — the key VRAM management mechanism.

Step	Time	Process	Output	Machine
1	8:00 PM	RSS scrape + dedup + timestamp alignment	Timestamped articles	M3
2	8:30 PM	FinBERT English scoring (FP16, batch=16)	EN sentiment CSV	M2
3	8:30 PM	MuRIL Hindi scoring (parallel to step 2)	HI sentiment CSV	M1
4	9:00 PM	Trust-weight fusion of both CSVs	Handshake CSV	M1
5	9:05 PM	T-LSTM inference on Handshake CSV	predictions.csv	M1

Hindi sparse rule: If a stock has fewer than 2 Hindi articles in the past 3 trading days, Hindi trust weight is set to 0.0 and English weight normalised to 1.0 for that stock on that day.

2.2 Stream B — LSTM Price History (Machine 1)

Input Features per Timestep

- Open, High, Low, Close (min-max normalised per stock)
- Volume (log-scaled)
- 5-day and 20-day moving average delta
- Realised volatility (5-day rolling std)
- RSI 14-period
- Market regime flag (0=normal, 1=moderate, 2=event)

Dynamic Window Sizing

- RealVol < 1.0x historical avg -> 60-day window
- RealVol 1.5-2.0x avg -> 30-day window
- RealVol > 2.0x avg -> 10-day window (event mode)
- Event triggers: RBI decisions, Union Budget, elections

Window recomputed fresh each day before inference.

2.3 Fusion Layer & Output Head (Machine 1)

Layer	Specification	Notes
Transformer output	512-dim embedding	From FinBERT / MuRIL

Layer	Specification	Notes
LSTM output	256-dim hidden state	Last timestep output only
Concatenation	768-dim fusion vector	Direct concat of both streams
FC Layer 1	768 -> 128, ReLU, Dropout 0.3	Learns cross-modal interactions
FC Layer 2 (output head)	128 -> 1, Linear (no activation)	Raw float output
Loss function	Huber loss (delta = 0.01)	Robust to crash-day outliers
Output	Float: predicted % move	e.g. +0.018 means predicted +1.8%
Confidence interval	MC Dropout, 50 forward passes	Mean +/- 1.96 * std = 95% CI

3. Hardware Cluster Allocation

Machine	Specs	VRAM Budget	Responsibilities
Machine 1 (Training Core)	RTX 5060, 8GB VRAM, 32GB RAM	~6.5 / 8 GB	MuRIL Hindi NLP, T-LSTM training, EWC weekly nudge, monthly full retrain, feature fusion, trust weights, Fisher matrix
Machine 2 (English NLP Hub)	RTX 3050, 4GB VRAM, 16GB RAM	~3.2 / 4 GB	FinBERT batch inference (FP16, batch=16), English sentiment CSV generation, embedding output to handshake
Machine 3 (Data Server)	RTX 3050, 4GB VRAM, 8GB RAM	~0.5 / 4 GB	RSS scraping, dedup, timestamp alignment, Kaggle OHLCV serving, 10AM price capture, data quality monitoring, cron scheduler

VRAM Optimisation Notes

- Machine 2: torch.cuda.amp.autocast() (FP16) + batch_size=16 keeps FinBERT under 3.5GB peak
- Machine 1: CSV handshake means BERT models never load during LSTM training -> saves ~3GB VRAM
- Monthly retrain: use mixed precision + gradient checkpointing if VRAM OOM as dataset grows
- All machines: use pathlib for all file paths. Zero hardcoded path strings allowed.
- Conda environment.yml: pin all versions. Test cold-start on all 3 machines before any PR merge.

4. Data Pipeline

4.1 Historical Price Data (Kaggle 20-Year Dataset)

Train / Validation / Holdout Split

- Years 1-17: Expanding walk-forward training set
- Years 18-19: 3-month rolling validation window
- Year 20: SEALED HOLDOUT -- never used for tuning

Walk-forward window expands 1 month per monthly retrain. Holdout unsealed only at final project evaluation.

Label Construction Formula

raw_ret = (price_10am_t1 - close_t0) / close_t0
adjusted_ret = raw_ret - nifty50_overnight_ret

Subtracting Nifty 50 overnight return removes global market drift (SGX Nifty, US market) to isolate the stock-specific news contribution. Label is a continuous float, not a class.

4.2 News Sources

Source	Language	Coverage	Fallback
Economic Times, Moneycontrol	English	All 15 stocks	LiveMint, Financial Express
Business Standard, NDTV Profit	English	All 15 stocks	Hindu BusinessLine
Navbharat Times, Amar Ujala	Hindi	Large-cap stocks	Dainik Bhaskar
Zee Business Hindi	Hindi	Financial-specific	NDTV Profit Hindi
NSE Official Announcements	English	All 15 stocks (authoritative)	BSE Announcements

5. Three-Tier Adaptive Learning System

Three independent feedback layers at different timescales. Tier 1 never modifies model weights. Tier 2 nudges the output head only. Tier 3 fully retrains with strict deployment gating.

TIER 1 -- Daily Source Trust Weight Update (10:10 AM every trading day)

- What changes: Trust score per news source per stock. NOT model weights.
- Logic: If prediction gap improved vs yesterday -> increase trust of sources that contributed
- Logic: If prediction gap worsened -> decrease trust of contributing sources
- Bounds: Trust weight hard-clamped to [0.1, 2.0]. No source can dominate or be silenced.
- Diversity guard: No single source may exceed 40% of total weight for any stock
- Anomaly guard: Skip update if |actual_move| > 3 sigma historical std (budget day, crash, etc.)
- Stale decay: Sources inactive 7+ days decay toward 1.0 neutral at 2% per day
- Storage: trust_weights/{date}.json -- immutable, versioned, committed to Git nightly

TIER 2 -- Weekly EWC Weight Nudge (Every Friday 11:00 PM)

- What changes: Last 2 FC layers of T-LSTM head ONLY. LSTM backbone and Transformer encoders frozen.
- Method: Elastic Weight Consolidation (EWC) -- Fisher matrix penalises overwriting critical weights
- Settings: Learning rate 1e-5 (10x smaller than base), 3 epochs, EWC lambda = 400
- Fisher matrix: Computed once after each monthly retrain. Reused for all nudges until next retrain.
- Rollback: If 7-day MAE after nudge > before nudge (+2% tolerance) -> restore previous checkpoint
- Skip condition: If any day that week triggered the anomaly guard -> skip entire nudge that week
- Purpose: Adapts to regime shifts without catastrophic forgetting of 20-year historical patterns

TIER 3 -- Monthly Full Walk-Forward Retrain (Last Sunday of each month)

- What changes: Full model retrained from scratch on expanding dataset (all history to current month)
- Holdout rule: Year 20 NEVER included in training or validation. Sealed until final evaluation only.
- Validation: New model evaluated on 3-month rolling window before deployment decision
- Deployment gate: New model MAE must beat live model MAE by >= 2% to replace the live model
- If gate fails: Live model remains active. New model archived for analysis.
- Post-deploy: Recompute Fisher Information Matrix on new training set for next months EWC nudges
- Duration: ~4-6 hours on Machine 1 RTX 5060. Runs Sunday to avoid weekday disruption.

EMERGENCY FREEZE -- Set FREEZE_ALL_UPDATES = True in config.yml

- Disables all three tiers simultaneously. Model still makes predictions but learns nothing.
- Use on: Union Budget day, election results, surprise RBI decisions, global market shocks
- Reset manually to False once market normalises -- typically 2-3 trading days later
- Rationale: Atypical event days break the news-to-price relationship. Learning from them corrupts state.

6. Output Specification

6.1 predictions.csv Daily Schema

Field	Type	Description	Example
date	DATE	Trading day of prediction	2026-03-15
ticker	STRING	NSE stock code	RELIANCE
predicted_pct	FLOAT	Predicted adjusted % move at 10AM	+0.018 (+1.8%)
conf_low / conf_high	FLOAT	95% confidence interval bounds	+0.014 / +0.022
direction	STRING	UP / DOWN / NEUTRAL (sign of pred)	UP
magnitude_label	STRING	Strong / Moderate / Weak (CI width)	Moderate
model_version	STRING	Checkpoint hash for reproducibility	v2.3-20260301
active_trust_weights	JSON	Source weights used this prediction	{ET:1.4, NBT:0.8}
actual_pct	FLOAT	Filled in at 10:05 AM next day	+0.021
gap	FLOAT	|pred_pct - actual_pct| (filled 10:05 AM)	0.003 (0.3%)

6.2 Team Dashboard Components

Daily View

- All 15 stocks: predicted % move + confidence interval bar
- Direction badge: UP / DOWN / NEUTRAL
- Magnitude label: Strong / Moderate / Weak
- Yesterday: actual vs predicted, gap highlighted
- Active model version + trust weight snapshot
- Anomaly flag / freeze status indicator

Trend View

- 7-day and 30-day rolling MAE per stock
- MAE slope chart -- is the gap shrinking?
- Source trust weight history per source
- Weekly nudge win / rollback rate over time
- Worst stock (highest MAE), best source
- Monthly retrain deploy / skip decision log

7. Evaluation Metrics & Success Criteria

Metric	Type	Definition	Stage 1	Stage 2
MAE (PRIMARY KPI)	Regression	Mean |pred_pct - actual_pct|	< 1.5%	< 0.8%
Direction Accuracy	Classification	sign(pred) == sign(actual)	> 52%	> 57%
High-Vol Accuracy	Classification	Dir. acc when |move| > 1.5%	> 50%	> 55%
Hybrid vs Baseline	Comparative	Hybrid MAE vs price-only LSTM	Baseline set	>=5% better
Trust Entropy	Health	Shannon entropy of trust weights	Monitor	> 60% max
Nudge Win Rate	Health	% weekly nudges that improved MAE	Monitor	> 60%
Gap Trend Slope	Health	Linear slope of 30-day rolling MAE	Monitor	Negative

Mandatory gate: Train price-only LSTM baseline in Stage 1. Record MAE. Hybrid must beat this by >= 5% to justify the full NLP infrastructure at Stage 2.

8. Staged Roadmap (1-3-36)

Stage	Timeline	Objectives	Exit Criteria
Stage 0	Weeks 1-2	Conda env, Git, pathlib. Kaggle data loaded. RSS feeds validated. 10AM alignment tested.	All 3 machines cold-start identically from environment.yml
Stage 1	Month 1	Pilot: 3 stocks. Price-only baseline. Label validation. MuRIL vs IndicBERT evaluation.	Baseline MAE recorded. Hindi NLP winner chosen.
Stage 2A	Months 2-4	Full T-LSTM built. CSV handshake live. Hybrid on 3 pilot stocks. Tier 1 daily trust updates.	Hybrid beats baseline by >= 5% on pilot stocks.
Stage 2B	Months 5-8	Scale to all 15 stocks. Tier 2 EWC live. Regime detector active. Dashboard v1.	All 15 stocks live. 30-day MAE slope negative.
Stage 2C	Months 9-12	Tier 3 monthly retrain automated. Full scheduler. Dashboard v2 with trend view.	Fully automated. No daily manual intervention needed.
Stage 3	Year 2-3	Full 20yr dataset. Holdout unsealed for final evaluation. Research documentation.	Final MAE vs holdout year reported.

9. Risk Register

Risk	Level	Mitigation
Data leakage via random train/test split	CRITICAL	Walk-forward CV only. Audit every split call in codebase before training.
SGX Nifty overnight confounding	HIGH	Subtract Nifty 50 overnight return from every label. Non-negotiable.
Catastrophic forgetting in EWC nudge	HIGH	EWC lambda=400. Rollback gate (+2% MAE tolerance). Skip on anomaly weeks.
Trust weight collapse to one source	MEDIUM	Hard bounds [0.1,2.0]. No source > 40% of total weight per stock.
IndicBERT poor on Hinglish finance text	MEDIUM	Evaluate MuRIL first in Stage 0. Fine-tune on 500 financial Hindi samples if needed.
FinBERT VRAM OOM on Machine 2	MEDIUM	FP16 + batch=16 + 512-token truncation. CPU offload as fallback.
Monthly retrain time grows too long	LOW	Mixed precision + checkpointing. Limit to 8 stocks if retrain exceeds 8 hours.

10. Immediate Next Actions -- Stage 0

Complete All 7 Actions Before Writing Any Model Code

- ACTION 1: Create Conda environment.yml with pinned versions (torch, transformers, pandas, ta-lib, yfinance). Test cold-start on all 3 machines.
- ACTION 2: Load Kaggle 20-year Nifty 50 OHLCV dataset on Machine 3 SSD. Audit for data gaps, missing columns, split errors.
- ACTION 3: Validate RSS coverage -- confirm 2+ Hindi and 2+ English sources for each of 3 pilot stocks.
- ACTION 4: Evaluate MuRIL vs IndicBERT on 50 sample financial Hindi headlines. Record which handles code-switching (Hinglish) better.
- ACTION 5: Test 8 PM to 10 AM timestamp alignment on 5 real trading days including one with a market holiday.
- ACTION 6: Build confound-corrected label function. Spot-check on 10 known high-news days from historical Kaggle data.
- ACTION 7: Train price-only LSTM baseline on 3 pilot stocks. Record the MAE. This number is the permanent benchmark.

Blueprint v1.0 -- Architecture Complete -- March 2026